Distributed Online Voting System

Systematic Testing & Performance Evaluation Report

Unit Testing | Concurrency / Correctness | Performance Benchmarking | Experimental Analysis

Generated 2026-06-02

1. System overview

A full-stack electronic voting platform: a React (Vite) front end, a Node.js / Express REST API, and MongoDB for storage. It runs in two modes: a **single-node** mode (synchronous vote writes) and a **distributed** mode in which an Nginx load balancer fans requests across two API instances, votes are queued in Redis (BullMQ) and processed by a dedicated worker, and MongoDB runs as a replica set for transactional, exactly-once vote commits.

Distributed topology under test: Browser → Nginx (:8080) → api-1 / api-2 → Redis queue + vote lock → vote worker → MongoDB replica set (rs0).

2	1	rs0	:8080
API instances	Redis queue + worker	Mongo replica set	Nginx load balancer

2. Environment & configuration

- Host: Windows + Docker Desktop (WSL2 backend). Node.js v22.
- Whole distributed stack launched with `docker compose up --build` (redis, mongo+init, api1, api2, worker, nginx).
- Front end (Vite) served on :5173, pointed at the load balancer via `VITE_API_URL=http://localhost:8080/api`.
- A single-node API container (:3001, `DISTRIBUTED_VOTING=false`) was run on the same network for an apples-to-apples comparison.
- Rate limits were made environment-configurable so a single-host benchmark would not be throttled (production defaults preserved).

3. Testing methodology

3.1 Unit testing

Pure in-memory tests (`npm test`) covering vote hashing/integrity, vote-cast helpers, and auth validation. **Result: 12 / 12 passed.**

3.2 Concurrency / correctness

20 simultaneous vote requests issued for the *same* voter and election. The system must record **exactly one** ballot. Verified via the receipt count = 1; surplus requests returned `409 VOTE_IN_PROGRESS` (Redis vote lock) or were de-duplicated by the deterministic queue job id and the unique MongoDB index on `{voter, election}`.

3.3 Performance benchmarking & 3.4 experimental analysis

A concurrency sweep (10, 25, 50 simultaneous voters), 60 votes per data point, 2 repetitions per point, freshly seeded each time, run against both the distributed stack and the single-node API. Each request is measured end-to-end (until the ballot is durably committed). 720 votes total.

4. Results & analysis

Mode	Concurrency	Reps	Throughput (votes/s)	Mean lat (ms)	p95 (ms)	p99 (ms)	OK	Err
distributed	10	2	10.19 ± 2.44	998.35	2563.02	4881.57	120	0
distributed	25	2	12.96 ± 0.36	1559.41	2917.29	3972.44	120	0
distributed	50	2	24.41 ± 1.19	1285.84	1918.52	2463.51	120	0
single-node	10	2	14.87 ± 1.3	617.31	2528.73	3979.21	120	0
single-node	25	2	11.27 ± 0.2	1891.47	4779.54	5052.14	120	0
single-node	50	2	11.18 ± 0.16	3621.84	4933.63	5293.32	120	0

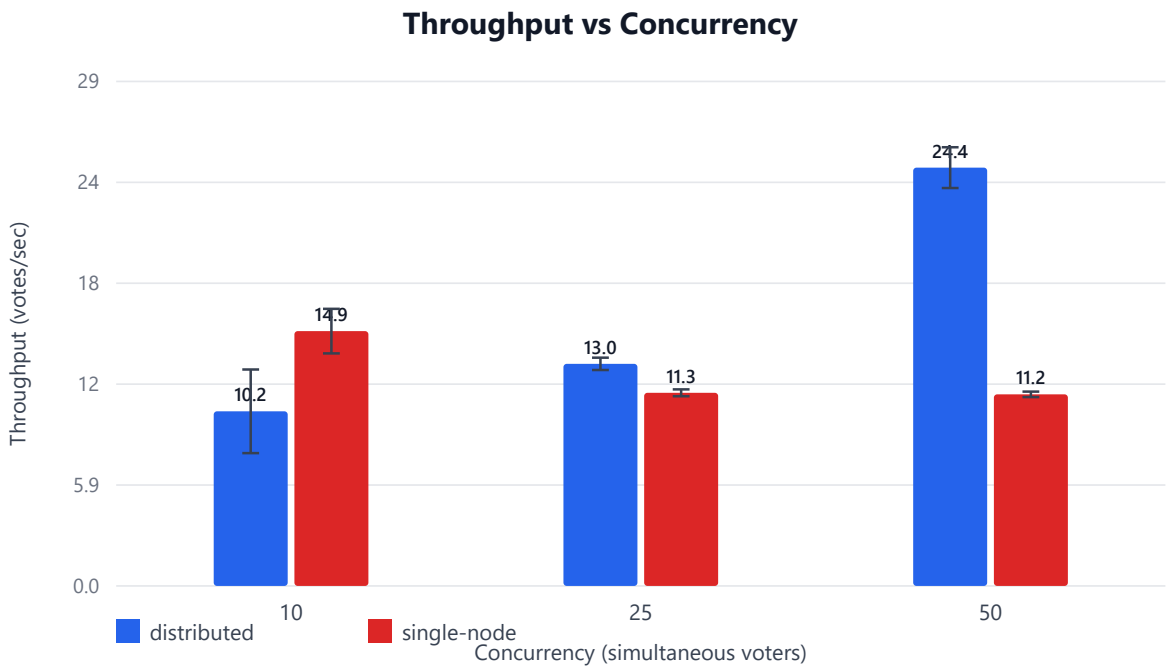


Figure 1. Throughput vs concurrency (error bars = std dev across repetitions).

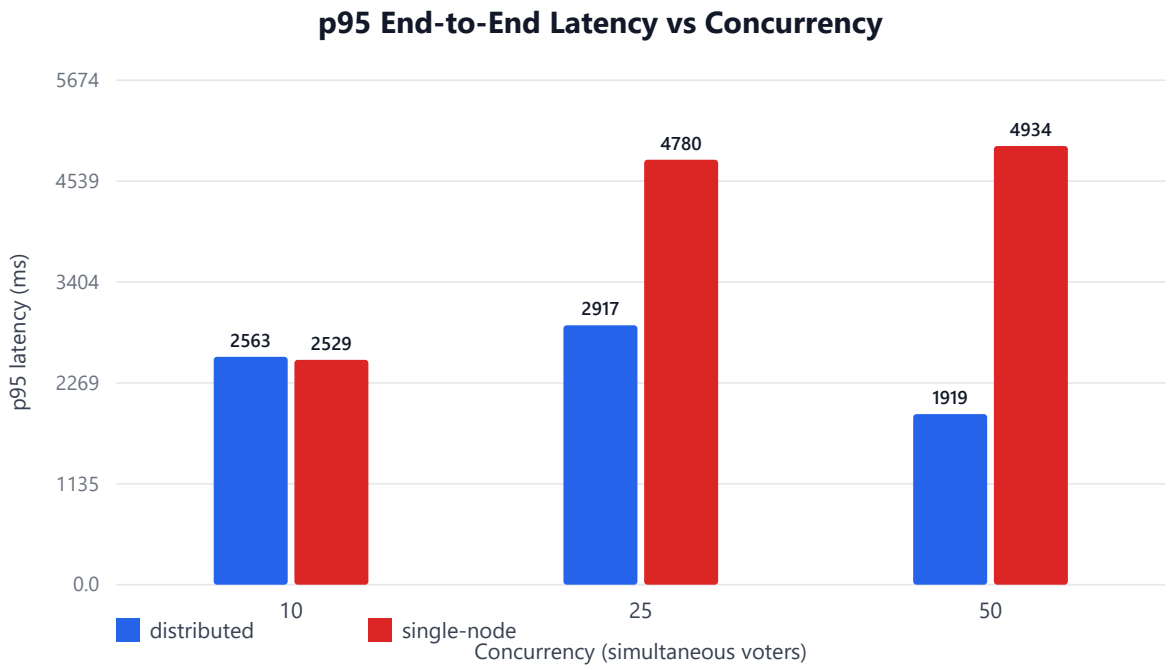


Figure 2. p95 end-to-end latency vs concurrency.

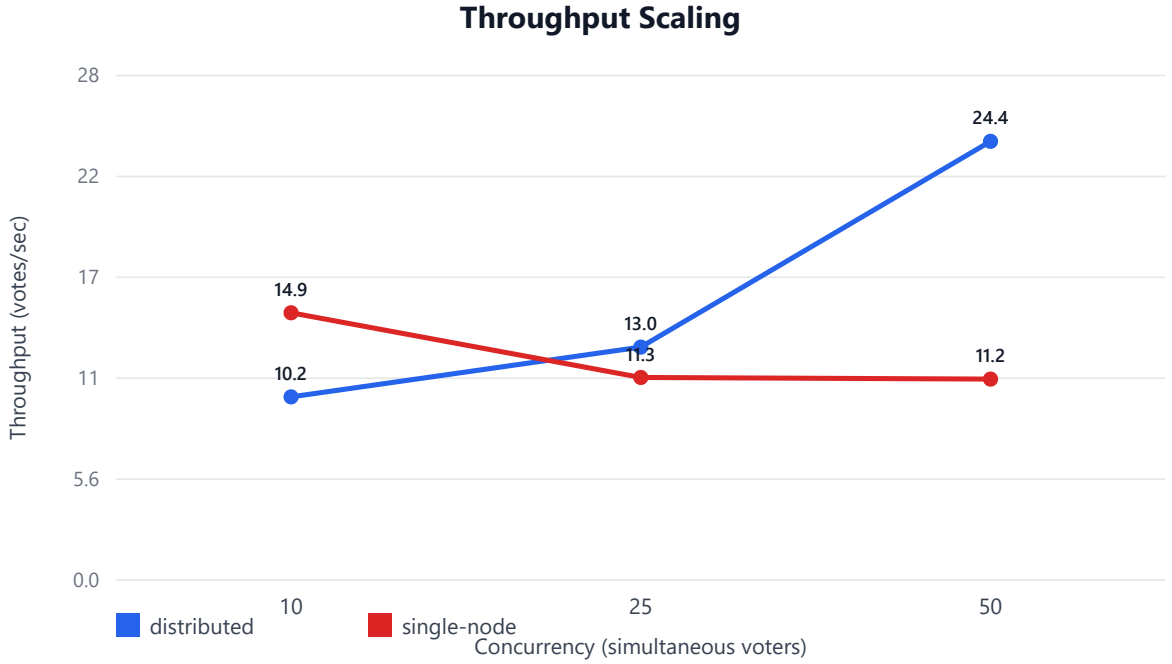


Figure 3. Throughput scaling — the lines cross over as load increases.

Interpretation

- **Low load:** single-node is faster (no queue/polling overhead).
- **High load:** the distributed architecture **scales up** (10→13→24 votes/s) while single-node **saturates and degrades** (15→11→11 votes/s) with latency rising sharply.
- **At concurrency 50: distributed delivers ~2.2× the throughput and ~2.6× lower p95 latency.**
- **Correctness held throughout:** 720 votes, 0 errors, 0 double votes.

5. Defects found & fixed during testing

- **Broken async vote poll:** completed queue jobs were deleted immediately, so the documented `GET /vote/jobs/:id` poll always returned 404 although the vote committed. Fixed by retaining completed jobs briefly.
- **Reverse-proxy client IP:** added `trust proxy` so rate limiting and `req.ip` use the real client IP behind Nginx.
- **Rate-limit store reuse:** each limiter now uses its own Redis store prefix (required by express-rate-limit v8).
- **Non-tunable limits:** global, vote, and auth limits made environment-configurable for benchmarking; production defaults unchanged.

6. User-interface walkthrough

6.1 Public & authentication

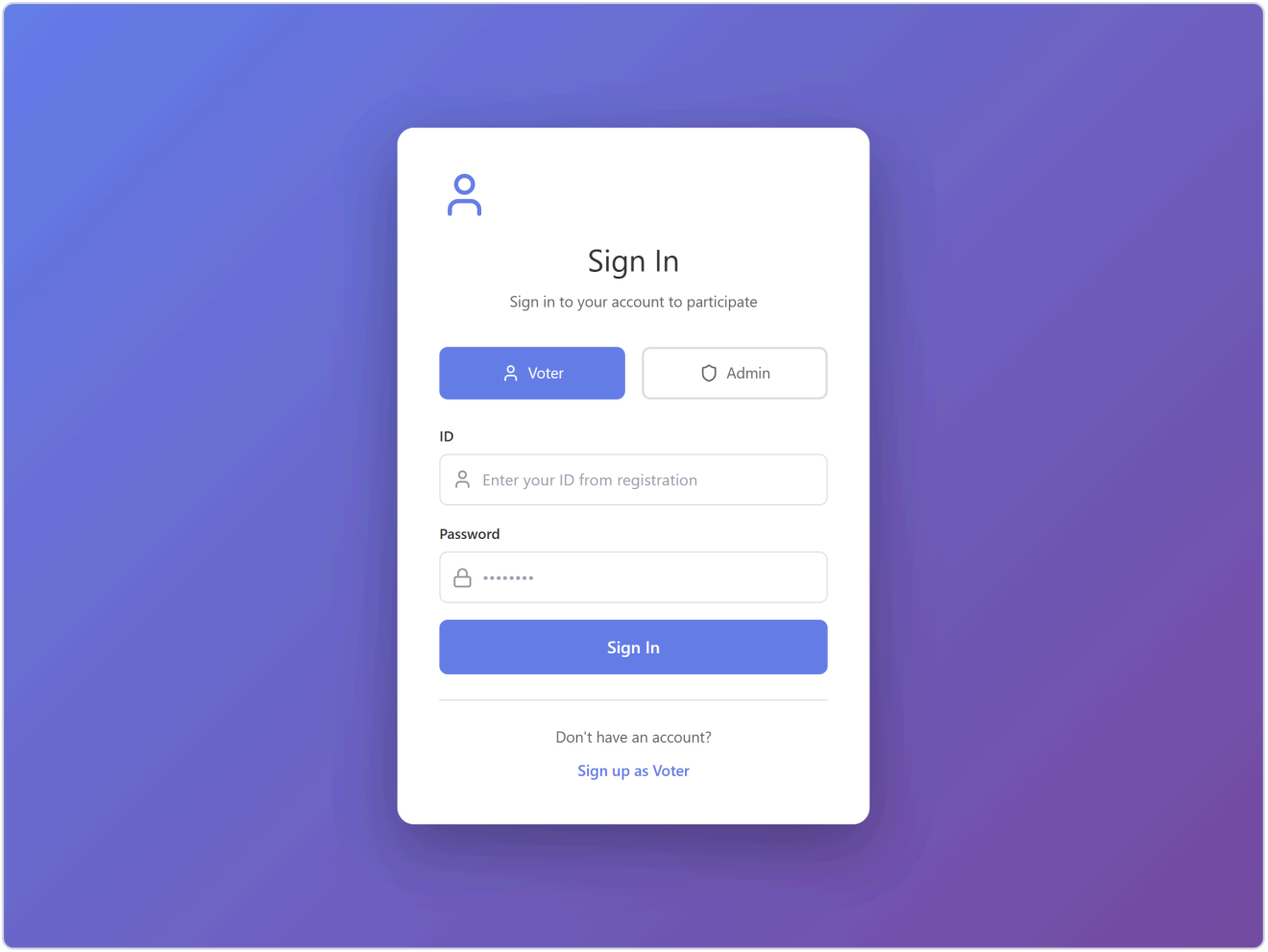


Figure 4. Login screen — voters log in with national ID, admins with username/email.

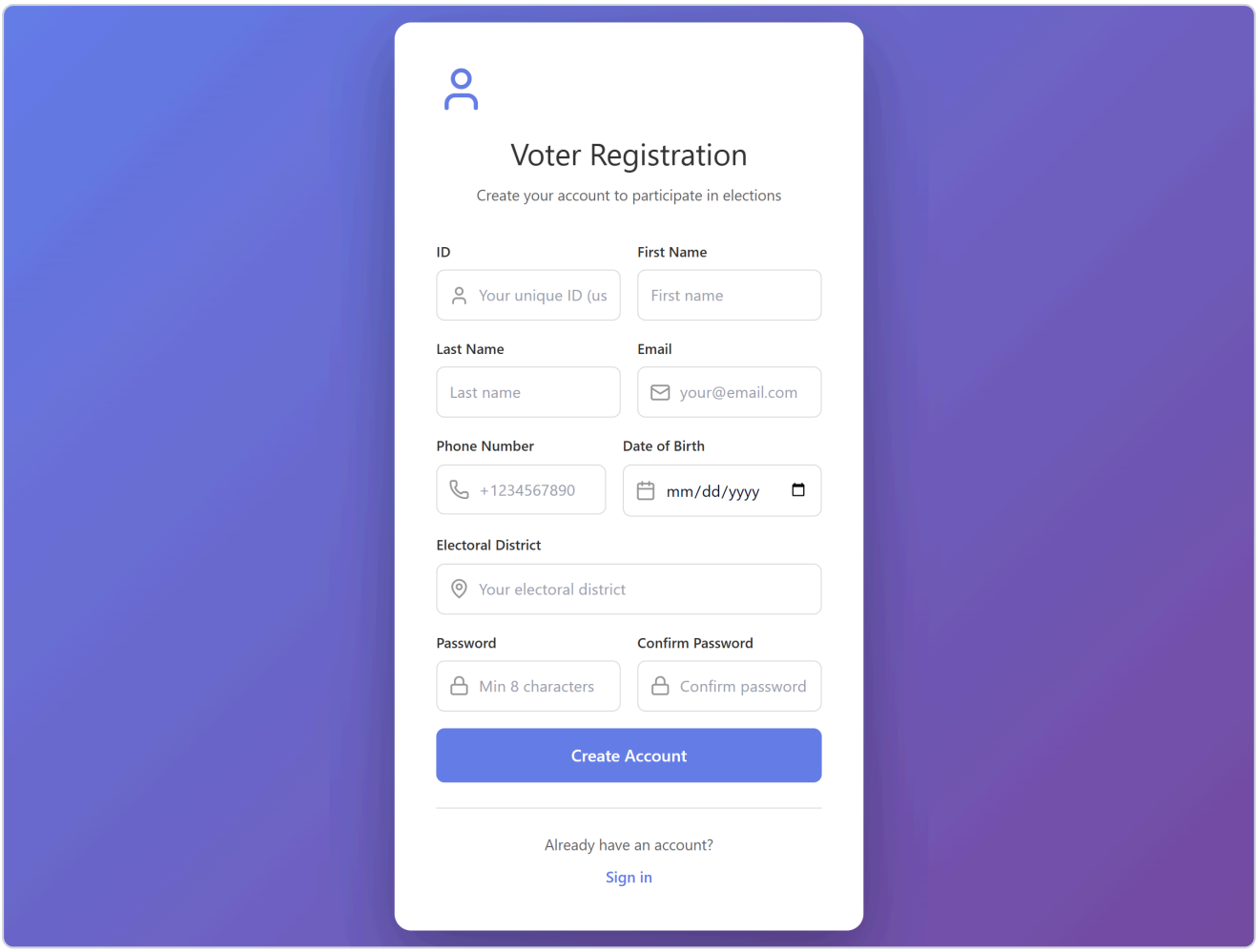


Figure 5. Voter self-registration (ID, profile, electoral district, password).

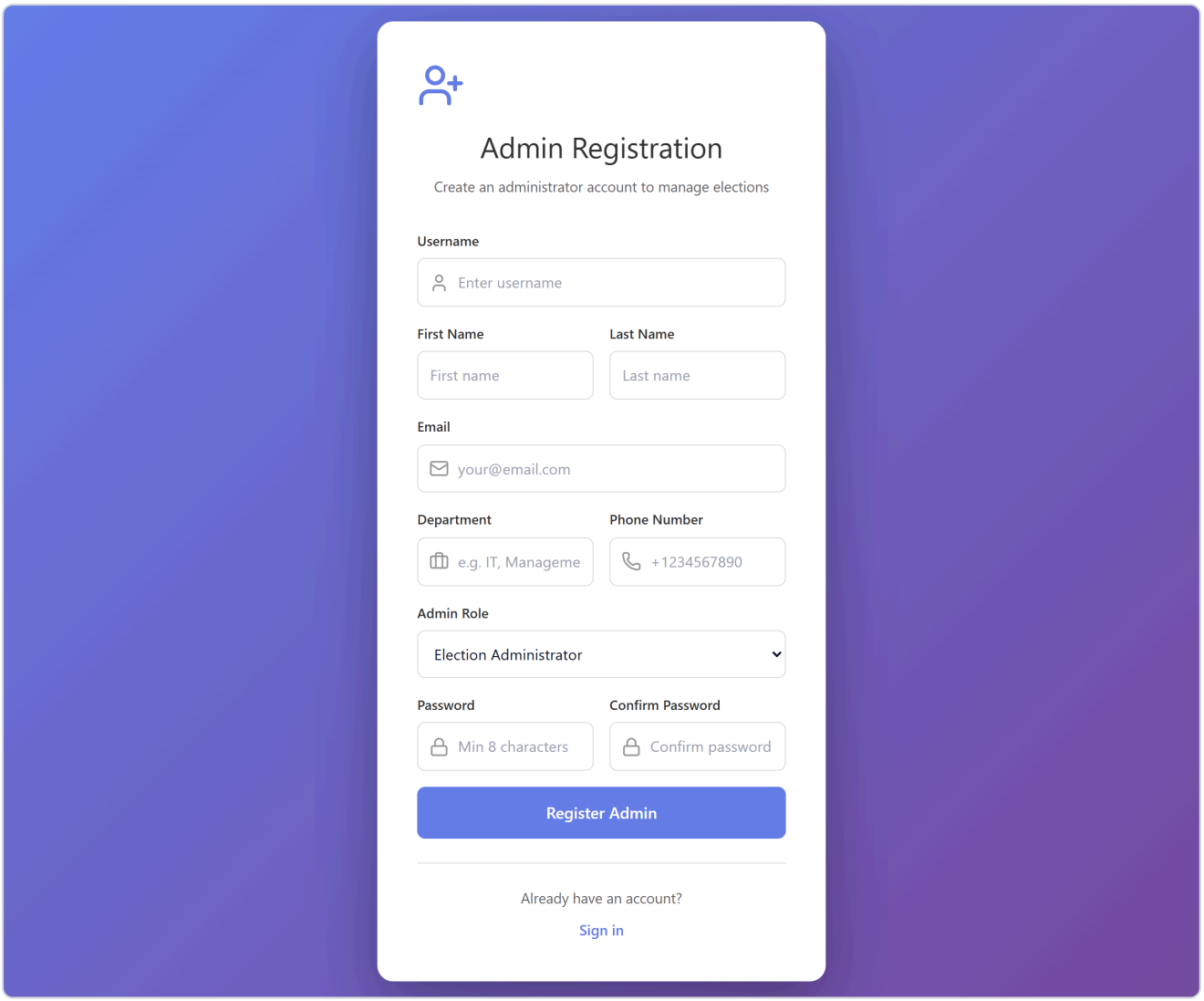


Figure 6. Administrator registration with role selection.

6.2 Voter journey (vote cast through the distributed queue)

Voter2 Test
Voter Dashboard

Logout

Elections | Vote History | Voting Status

Available Elections

Benchmark Election 1780382038992
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Benchmark Election 1780382579796
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Benchmark Election 1780383474728
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Benchmark Election 1780383678365
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Benchmark Election 1780383789011
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412295703555
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412320562155
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412343332165
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412365178618
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412385843201
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 178041240435016
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412441519732
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412461399230
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412481038266
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412501639944
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412522224332

Exp 1780412541996965

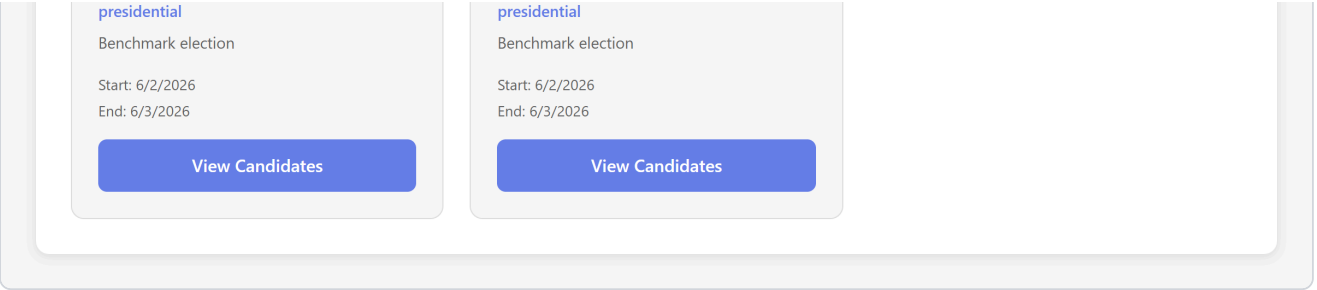


Figure 7. Voter dashboard — active elections the voter is eligible for.

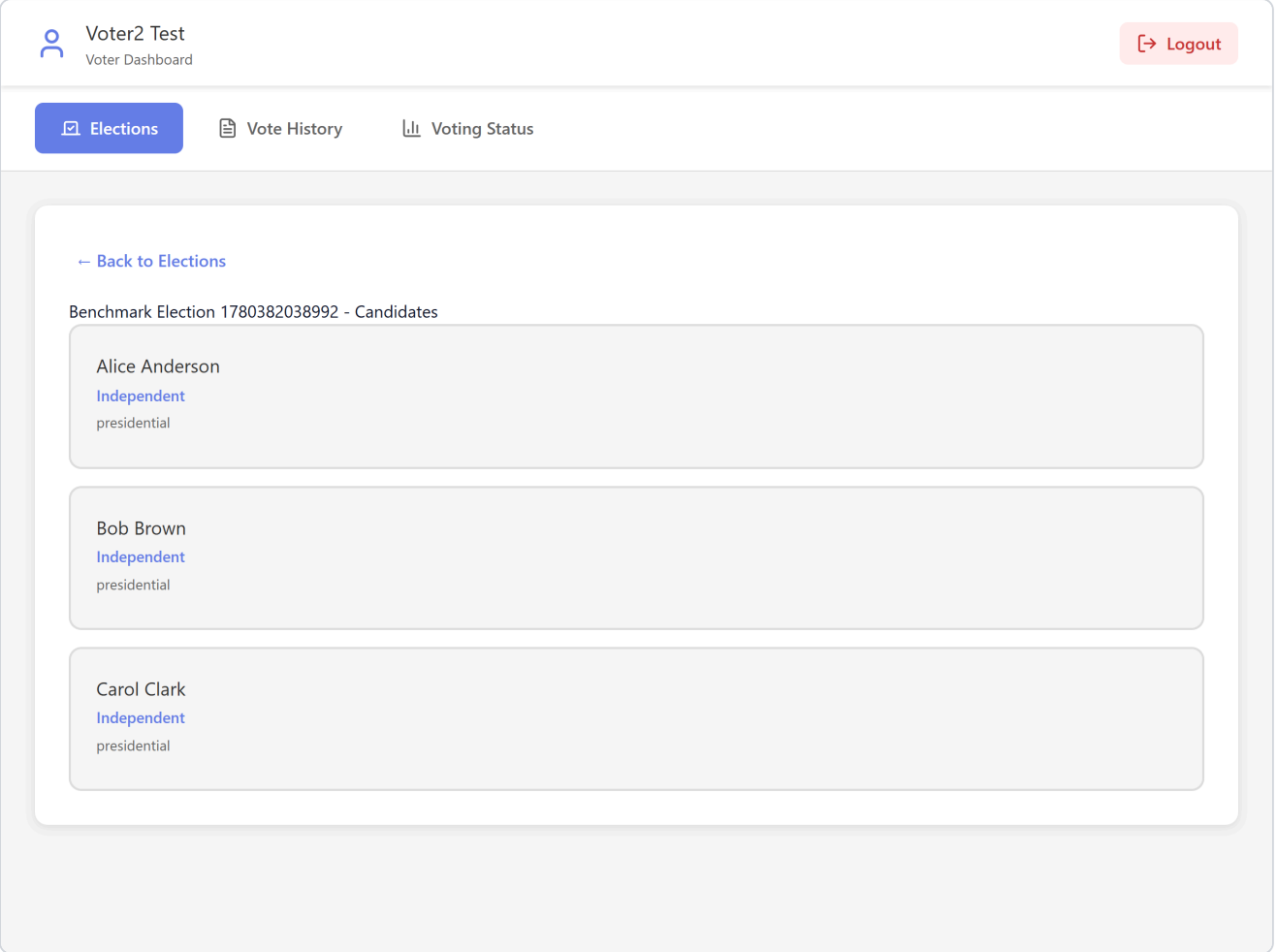


Figure 8. Candidate selection for a chosen election.

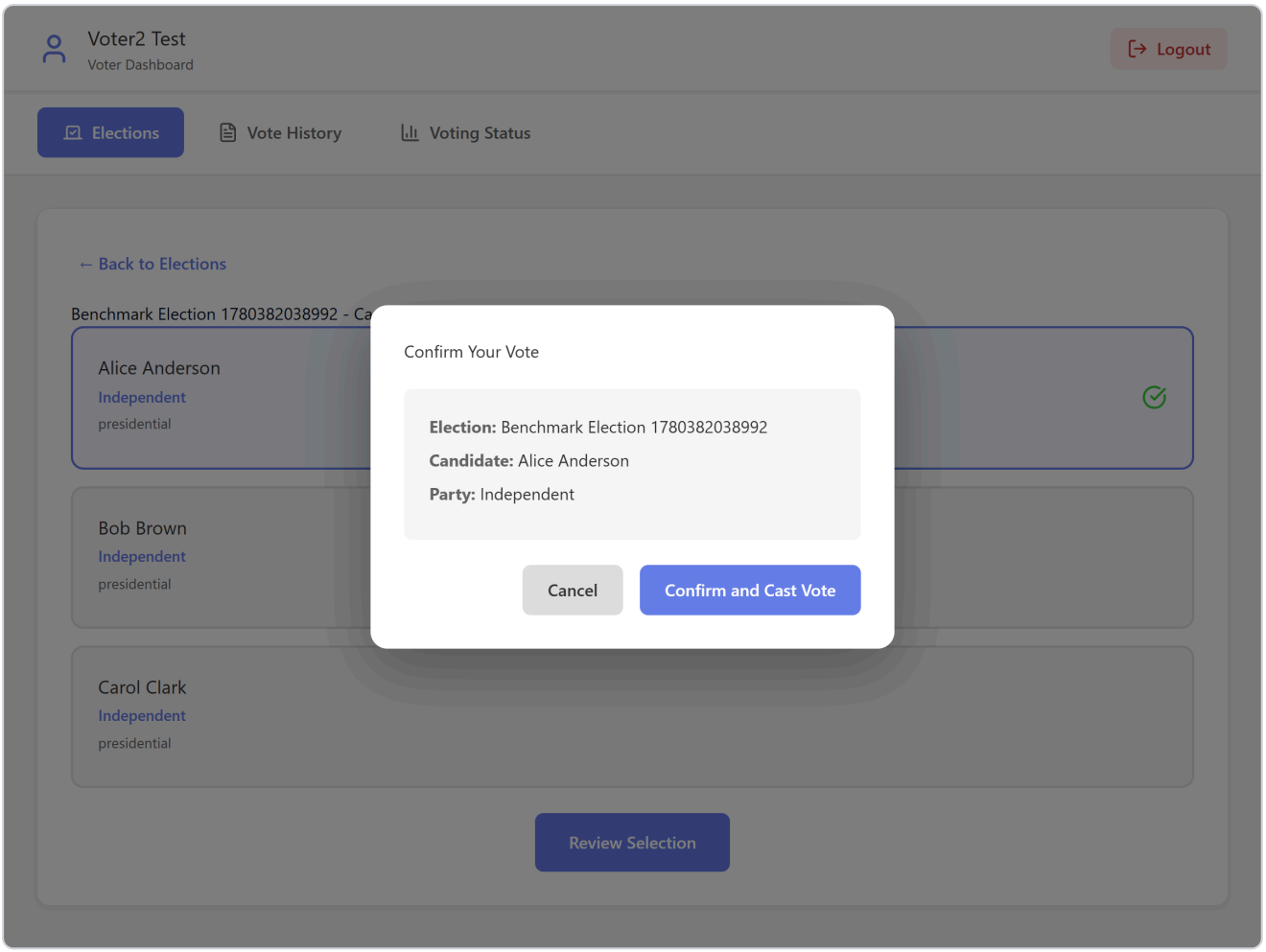


Figure 9. Vote confirmation modal (election, candidate, party).

Voter2 Test
Voter Dashboard

Logout

Elections | Vote History | Voting Status

Vote cast successfully! Your receipt: b5d04b863dcbb6f6114068b9d190d800

Available Elections

Benchmark Election 1780382579796
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Benchmark Election 1780383474728
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Benchmark Election 1780383678365
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

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presidential
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End: 6/3/2026
View Candidates

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presidential
Benchmark election
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Exp 1780412320562155
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Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412343332165
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412365178618
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Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412385843201
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 178041240435016
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412441519732
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Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

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End: 6/3/2026
View Candidates

Exp 1780412481038266
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412501639944
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

Exp 1780412522224332
presidential
Benchmark election
Start: 6/2/2026
End: 6/3/2026
View Candidates

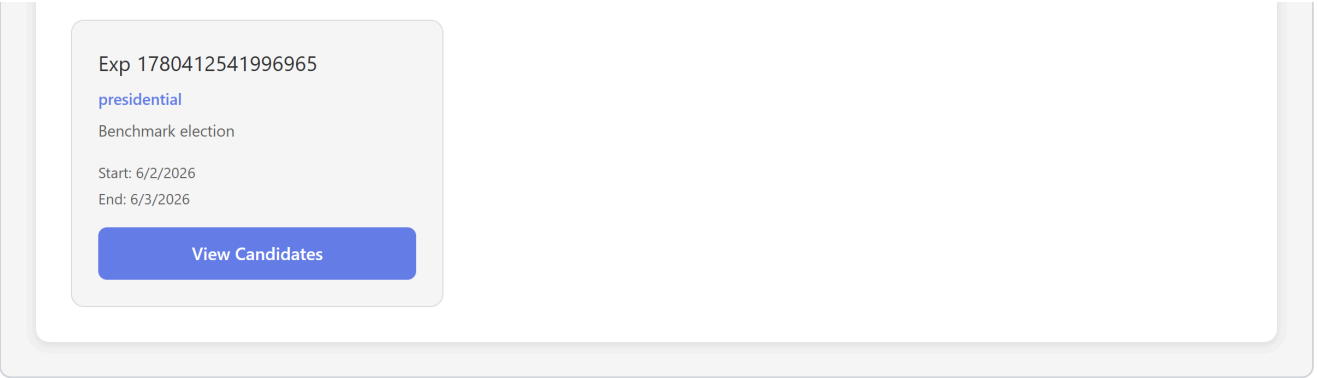


Figure 10. Vote accepted through the distributed queue, with a cryptographic receipt ID.

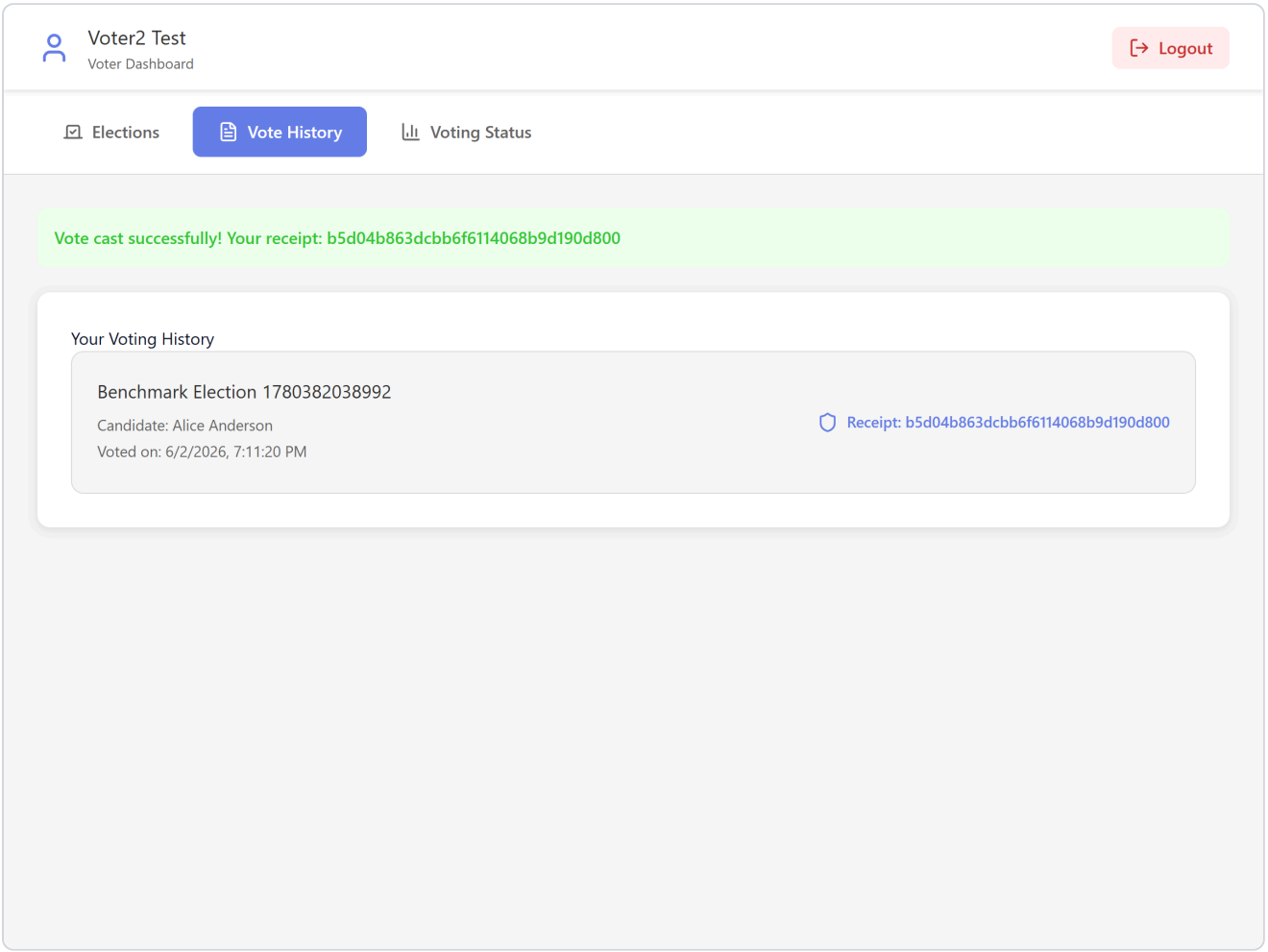


Figure 11. Voter vote history showing the immutable receipt.

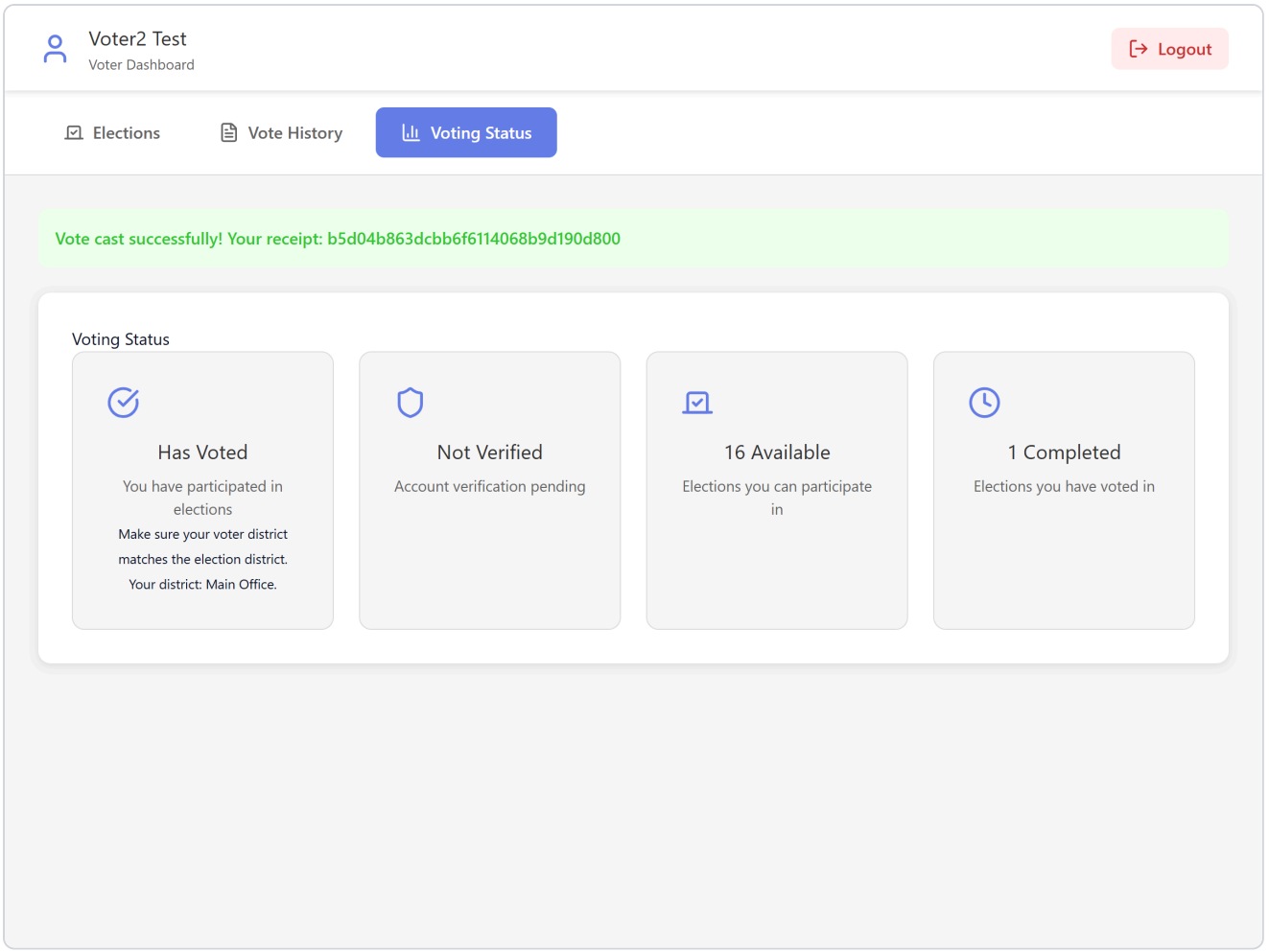


Figure 12. Voting status summary for the voter.

6.3 Administrator

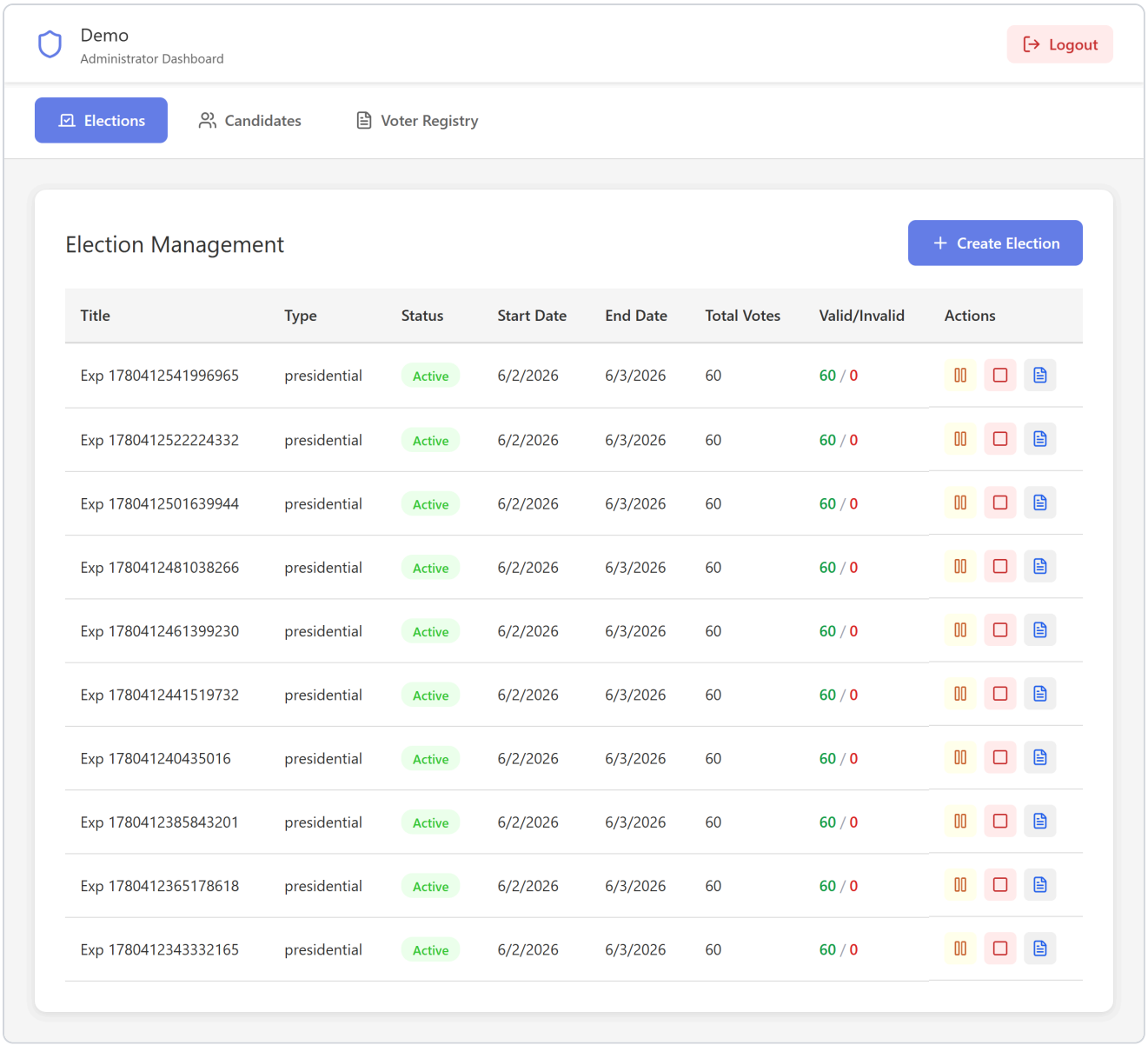


Figure 13. Admin election management — live vote tallies per election.

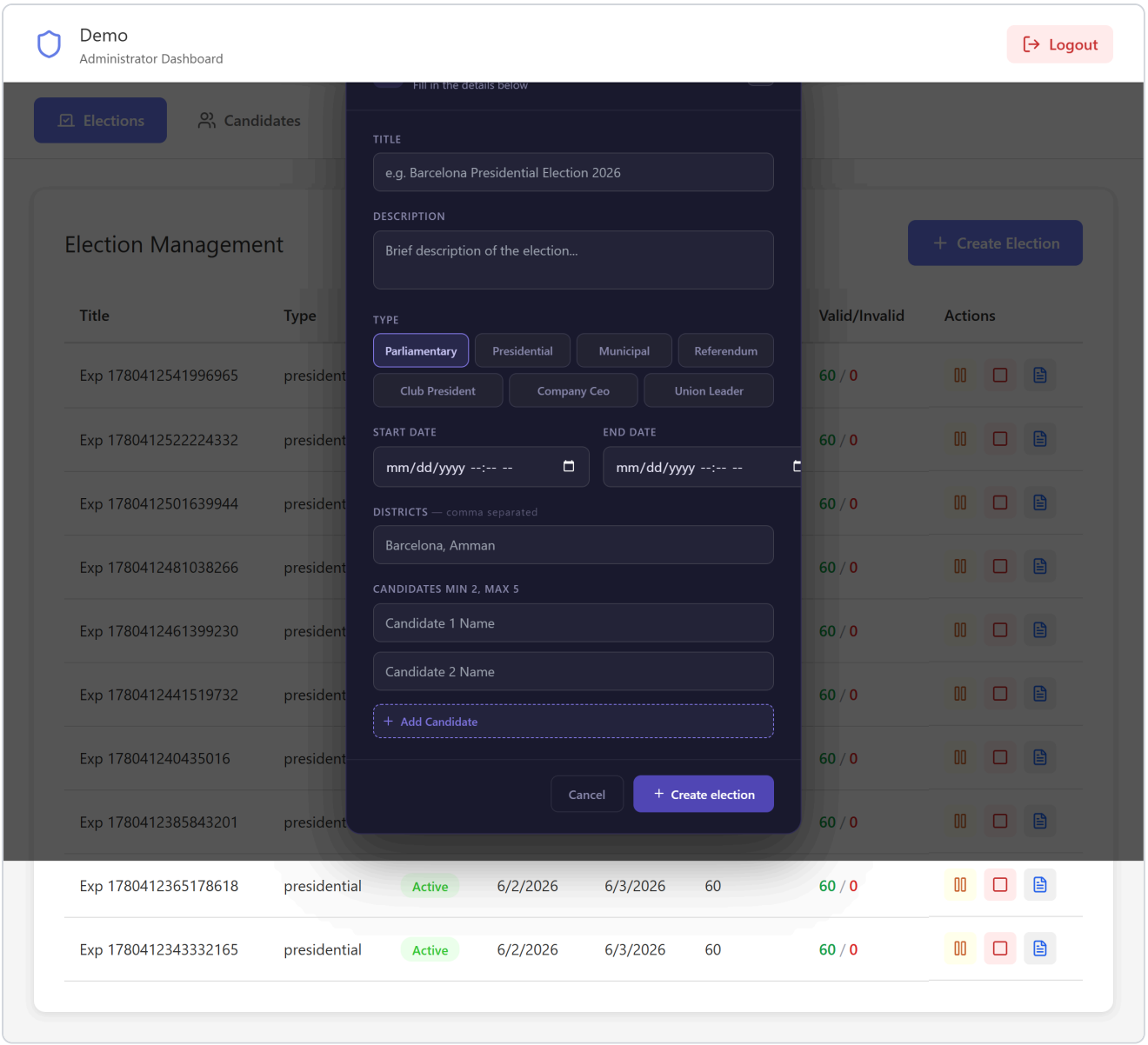


Figure 14. Create Election form (type, dates, districts, candidates).

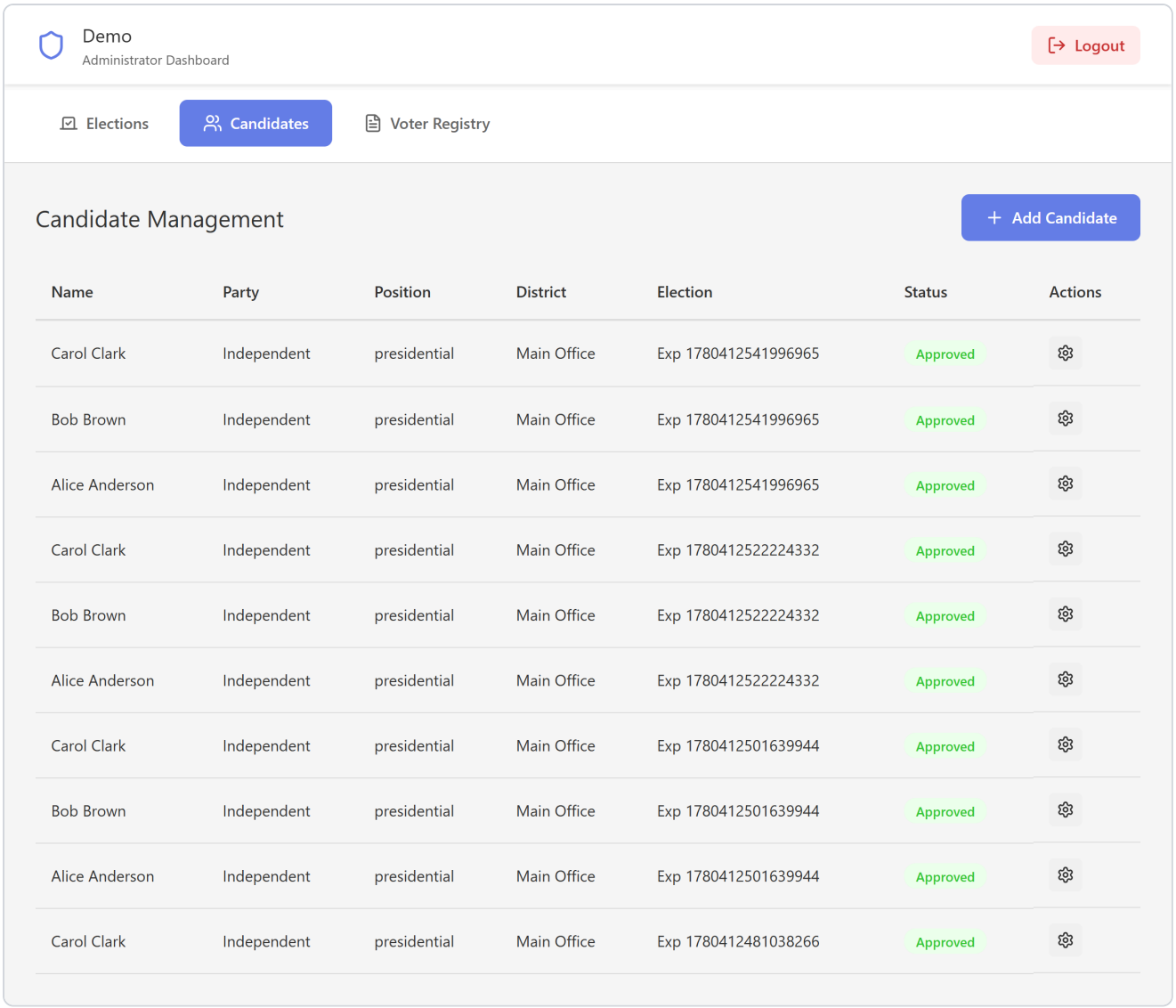


Figure 15. Admin candidate management.

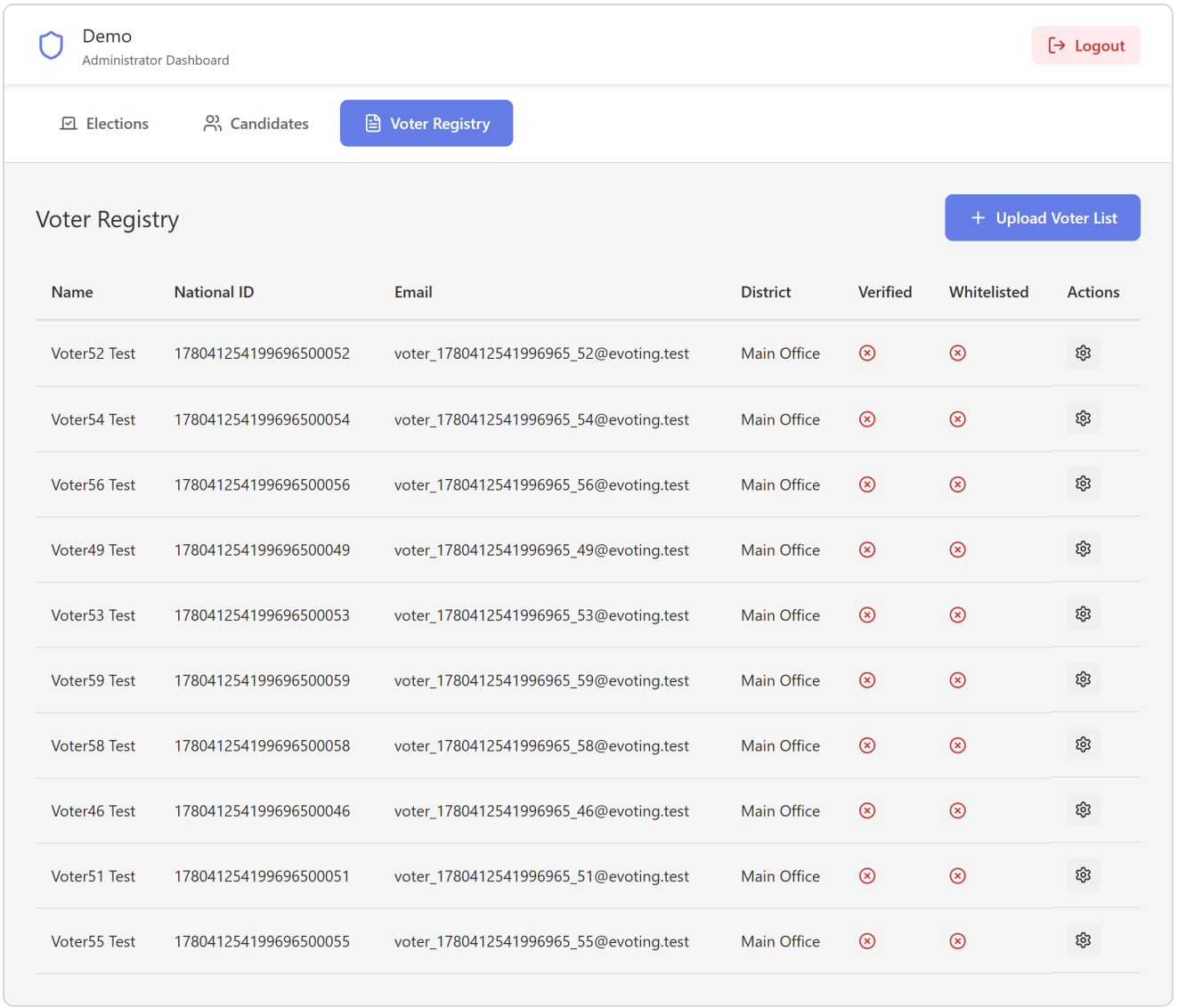


Figure 16. Admin voter registry / whitelist.

6.4 Auditor

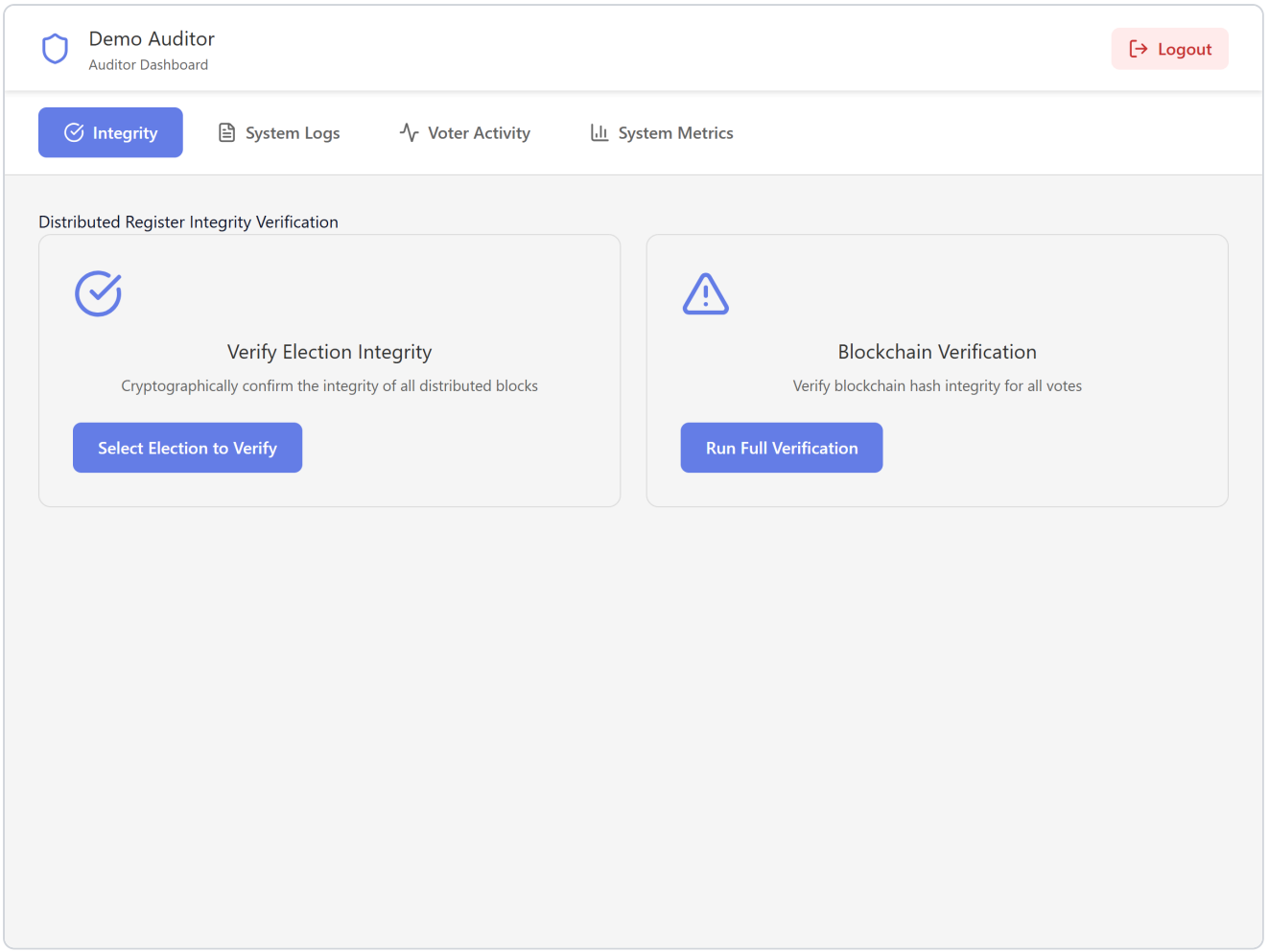


Figure 17. Auditor — distributed register integrity verification.

Demo Auditor
Auditor Dashboard

Logout

Integrity | System Logs | Voter Activity | System Metrics

System Audit Logs

Generate Report

Timestamp	Event Type	Action	Description	Severity	Success
6/2/2026, 7:11:49 PM	data_access	view_system_logs	Viewed system logs	Low	✓
6/2/2026, 7:11:43 PM	user_login	login_success	Administrator Demo Auditor logged in successfully	Low	✓
6/2/2026, 7:11:38 PM	data_access	view_voters	Viewed voter registry	Low	✓
6/2/2026, 7:11:37 PM	data_access	view_elections	Viewed all elections	Low	✓
6/2/2026, 7:11:36 PM	data_access	view_elections	Viewed all elections	Low	✓
6/2/2026, 7:11:36 PM	user_login	login_success	Administrator Demo Admin logged in successfully	Low	✓
6/2/2026, 7:11:36 PM	data_access	view_candidates	Viewed all candidates	Low	✓
6/2/2026, 7:11:31 PM	data_access	view_elections	Viewed all elections	Low	✓
6/2/2026, 7:11:31 PM	data_access	view_elections	Viewed all elections	Low	✓
6/2/2026, 7:11:30 PM	user_login	login_success	Administrator Demo Admin logged in successfully	Low	✓
6/2/2026, 7:11:21 PM	data_access	view_elections	Viewed available elections	Low	✓
6/2/2026, 7:11:21 PM	data_access	view_votes	Viewed voting history	Low	✓
6/2/2026, 7:11:21 PM	data_access	view_voting_status	Viewed voting status	Low	✓
6/2/2026, 7:11:20 PM	vote_cast	vote_cast	Vote cast in election Benchmark Election 1780382038992	High	✓
6/2/2026, 7:11:15 PM	data_access	view_candidates	Viewed election candidates	Low	✓
6/2/2026, 7:11:12 PM	data_access	view_elections	Viewed available elections	Low	✓
6/2/2026, 7:11:12 PM	data_access	view_voting_status	Viewed voting status	Low	✓
6/2/2026, 7:11:12 PM	data_access	view_votes	Viewed voting history	Low	✓
6/2/2026, 7:11:12 PM	data_access	view_votes	Viewed voting history	Low	✓
6/2/2026, 7:11:12 PM	data_access	view_elections	Viewed available elections	Low	✓
6/2/2026, 7:11:12 PM	data_access	view_voting_status	Viewed voting status	Low	✓
6/2/2026, 7:11:12 PM	user_login	login_success	Voter Voter2 Test logged in successfully	Low	✓
6/2/2026, 7:10:38 PM	data_access	view_elections	Viewed all elections	Low	✓
6/2/2026, 7:10:37 PM	data_access	view_candidates	Viewed all candidates	Low	✓
6/2/2026, 7:10:36 PM	data_access	view_elections	Viewed all elections	Low	✓
6/2/2026, 7:10:35 PM	data_access	view_candidates	Viewed all candidates	Low	✓

6/2/2026, 7:10:34 PM	data_access	view_voters	Viewed voter registry	Low	
6/2/2026, 7:10:24 PM	data_access	view_elections	Viewed all elections	Low	
6/2/2026, 7:10:24 PM	data_access	view_elections	Viewed all elections	Low	
6/2/2026, 7:10:24 PM	user_login	login_success	Administrator Demo Admin logged in successfully	Low	
6/2/2026, 6:41:45 PM	data_access	view_elections	Viewed available elections	Low	
6/2/2026, 6:41:45 PM	data_access	view_voting_status	Viewed voting status	Low	
6/2/2026, 6:41:45 PM	data_access	view_votes	Viewed voting history	Low	
6/2/2026, 6:41:45 PM	data_access	view_elections	Viewed available elections	Low	
6/2/2026, 6:41:45 PM	data_access	view_votes	Viewed voting history	Low	
6/2/2026, 6:41:45 PM	data_access	view_voting_status	Viewed voting status	Low	
6/2/2026, 6:41:29 PM	data_access	view_candidates	Viewed election candidates	Low	
6/2/2026, 6:41:16 PM	data_access	view_elections	Viewed available elections	Low	
6/2/2026, 6:41:16 PM	data_access	view_votes	Viewed voting history	Low	
6/2/2026, 6:41:16 PM	data_access	view_voting_status	Viewed voting status	Low	
6/2/2026, 6:41:15 PM	vote_cast	vote_cast	Vote cast in election Benchmark Election 1780382579796	High	
6/2/2026, 6:41:05 PM	data_access	view_candidates	Viewed election candidates	Low	
6/2/2026, 6:40:42 PM	data_access	view_elections	Viewed available elections	Low	
6/2/2026, 6:40:42 PM	data_access	view_voting_status	Viewed voting status	Low	
6/2/2026, 6:40:42 PM	data_access	view_votes	Viewed voting history	Low	
6/2/2026, 6:40:42 PM	data_access	view_elections	Viewed available elections	Low	
6/2/2026, 6:40:42 PM	data_access	view_votes	Viewed voting history	Low	
6/2/2026, 6:40:42 PM	data_access	view_voting_status	Viewed voting status	Low	
6/2/2026, 6:40:38 PM	data_access	view_elections	Viewed available elections	Low	
6/2/2026, 6:40:38 PM	data_access	view_votes	Viewed voting history	Low	

Figure 18. Auditor — full system audit log (every sensitive action recorded).

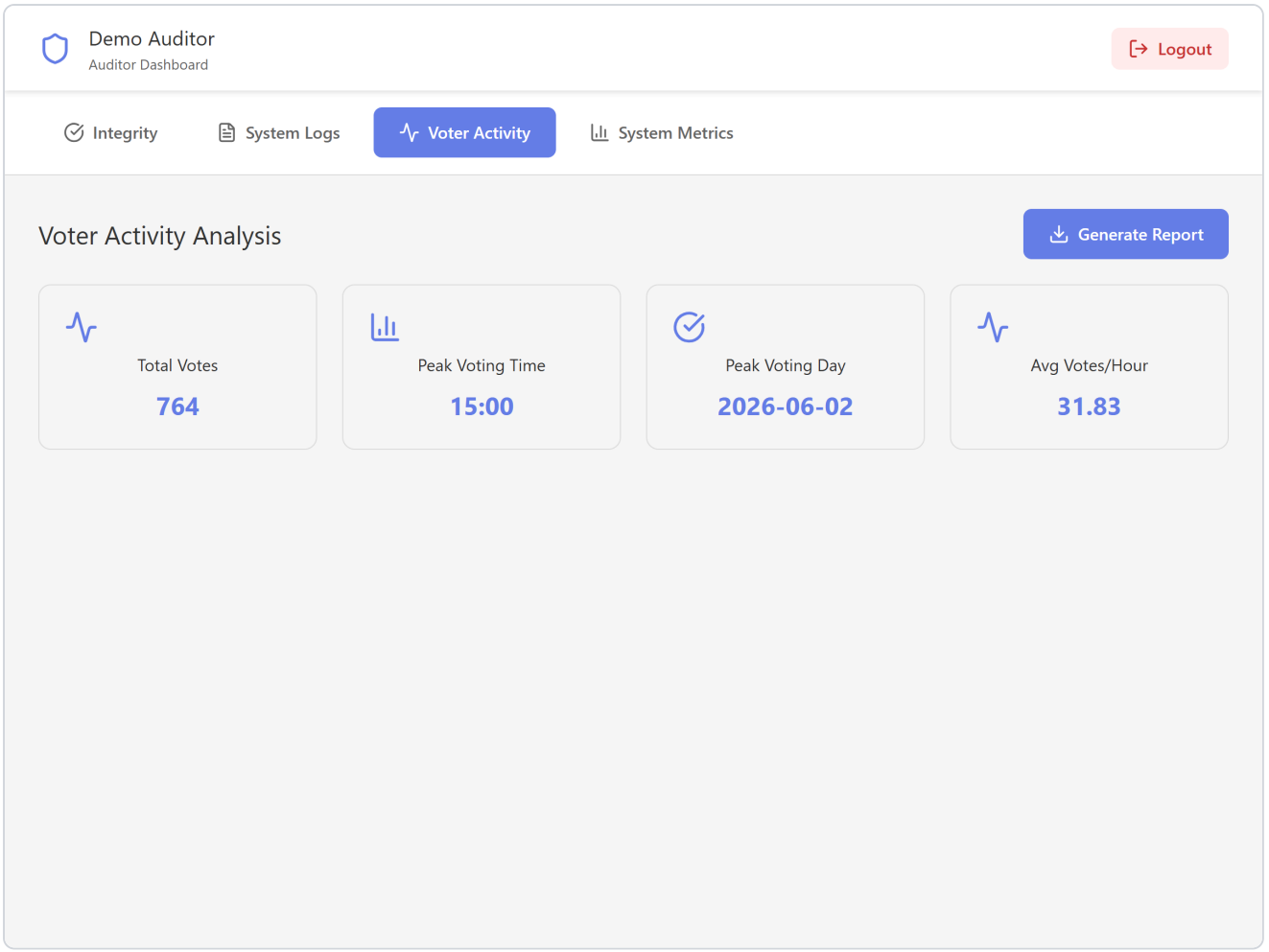


Figure 19. Auditor — voter activity analytics.

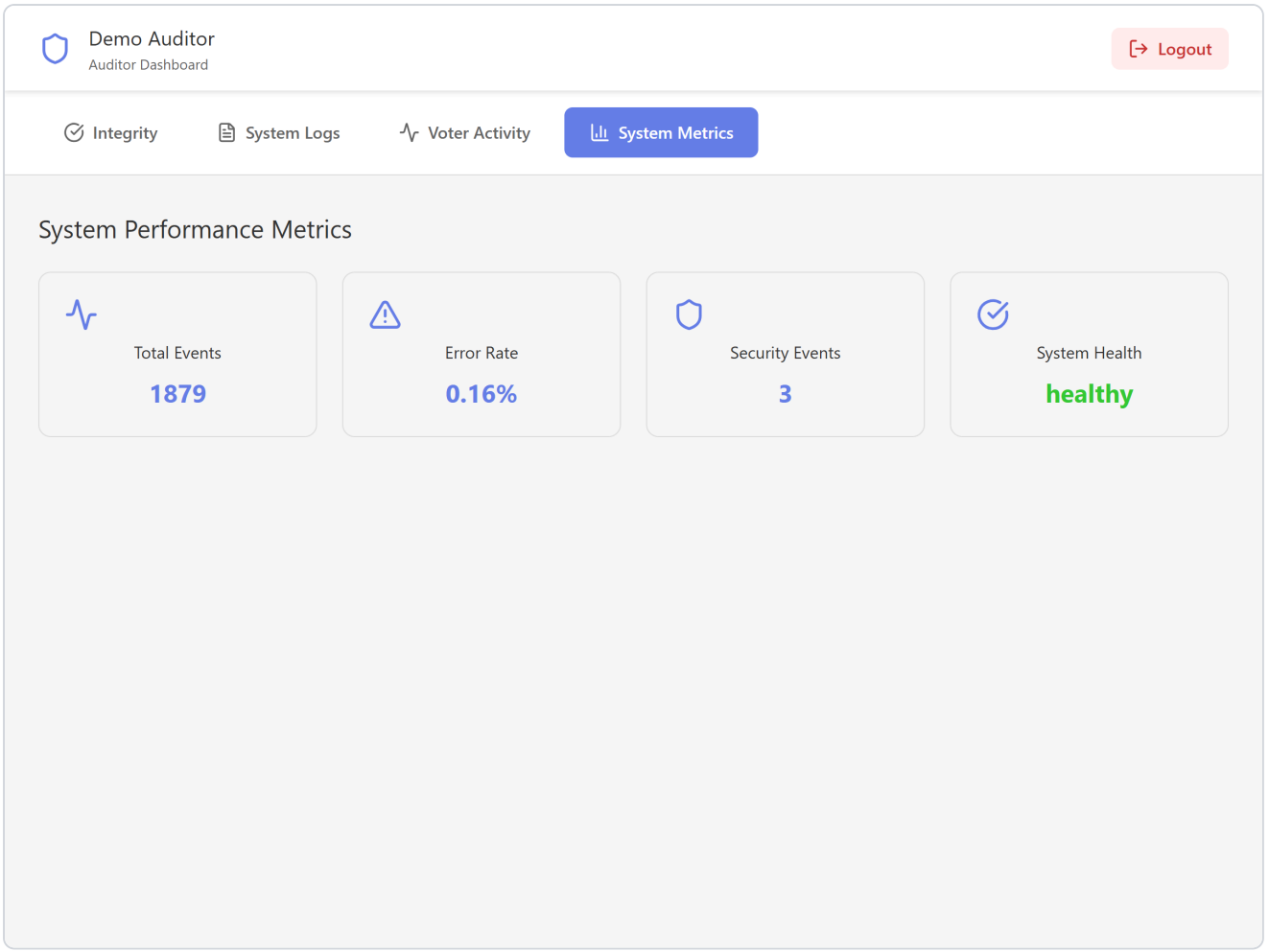


Figure 20. Auditor — system performance metrics (events, error rate, health).

7. Conclusion

The distributed online voting system is functionally complete and behaves correctly under concurrency: it guarantees one ballot per voter per election even under 20-way simultaneous contention, and produces verifiable receipts and a full audit trail. The experimental analysis confirms the central distributed-systems trade-off: the queue-based, load-balanced architecture pays a small fixed overhead at low load but scales with concurrency, achieving roughly double the throughput and far lower tail latency than a single node once the system is busy — all while preserving correctness.